

Learning objectives

In this lesson you will learn to:

- Define the term 'topology'
- Describe the characteristics of bus, star and mesh network topologies
- Draw and label a diagram of each topology

For more information and additional learner activities see Chapter 4.1 of the student book.

Our network

How do you think the school network is organised?

Do you know where the servers are?

Can you spot any network hardware in this room?



Organising a network

The way that a LAN is organised is known as its topology.
There are three topologies commonly used in LANs:

- Bus
- Star
- Mesh (partial and full).



Topology security

Bus topology

Not very secure as all devices 'see' all of the data traffic.

Star topology

Very secure as each transmission is sent directly to the intended recipient.

Mesh topology

Reasonably secure as the traffic takes the shortest route to its destination.

The internet

You have been focussing on topologies used for LANs so far. The internet is a WAN, but it is also an enormous mesh network.

The internet's long-distance, high-bandwidth cables that travel across the globe (often under the ocean) are known as the **backbone**. Each country generally has its own backbone, which can be linked to others using **Network Access Points** (NAPs)

Networks are connected to the internet using a **Point of Presence** (PoPs) provided by an **Internet Service Provider** (ISP). ISPs are organisations that 'rent' bandwidth from telecommunications companies such as BT and then charge their customers to use some of that bandwidth.

Wrap up: you have learned how to...

- ✓ Define the term 'topology'.
 - The physical organisation of a network.
- ✓ Describe the characteristics of bus, star and mesh network topologies.
- ✓ Draw and label a diagram of each topology.
- ✓ Match descriptions to network topologies.
- ✓ Recognise the role of components that make up the internet (backbone, POP, NAP, router).